

Soccerix: Software-based AI-powered Soccer Tactical Analysis Tool

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Chapter 1

Introduction

The project proposal for **Soccerix**, an AI-driven soccer analysis tool, outlines its development and deployment to revolutionize soccer tactical analysis through advanced data driven techniques. Soccerix will enhance team classification and analysis, incorporate object detection and tracking, and automate event detection for goals, free kicks, and corner kicks. Utilizing AI-based solutions and detailed statistics, the tool provides comprehensive post-match reports highlighting team strengths and weaknesses, offering a practical approach to performance improvement. This document provides a thorough overview of the problem addressed, the motivation behind the project, and the proposed solution.

1.1 Problem Domain

The problem domain is the tactical analysis of soccer matches, which traditionally made based on manual observation and subjective interpretation. Player movements, team formations and in-game incidents are usually assessed by coaches and analysts using their own eyes which is far from appropriate and very laborious. These traditional methods may overlook more subtle patterns commonly observed in team dynamics and strategic evolution, complicating the ability of the model fully capture the complexities of the game. Moreover, manual analysis is time consuming resulting in delayed post-match feedback, limiting the ability to make timely modifications. With soccer relying more on data-based approaches, teams that cannot instantly and accurately evaluate large volumes of match data are at a competitive disadvantage.

1.2 Research Problem Statement

The lack of an automated, impartial, and effective system for tactical football analysis is the research challenge. Conventional approaches, which rely on physical labour and human interpretation, are not consistent enough to provide a complete picture of team performance. Soccer's increasing focus on analytics has made an AI-driven solution more urgently necessary. Coaches and analysts could be able to make more educated tactical decisions with the help of such a system, which could offer precise, comprehensive post-match analysis. By examining how AI can provide more dependable, comprehensive, and timely insights on the tactical intricacies of the game, this research seeks to close the gap.

1.3 Software Requirements Specification

This section lays out the essential features and functions of Soccerix. The details are mentioned below.

1.3.1 Functional Requirements

1. Video Upload and Processing

The system shall allow users to upload a video file. This input video will be processed at the back-end to extract relevant frames for object, event, and field key-points detection, as well as for object tracking and other modules.

2. Dashboard Data Visualization

When the user navigates to the dash-board, then system shall show dynamic visualizations like ball trajectory, team pitch control diagrams and heatmaps.

3. Report Generation and Download

When the user clicks the "Process/Generate Report" button the system shall generate a comprehensive report summarizing the teams' tactical analysis. This report will be available for download as well.

1.3.2 Non-Functional Requirements

1. Usability

The system should have a easy to use interface with clear instructions for uploading video files.

2. Reliability and Availability

The system should work well all the time and be available for users to use without any problems.

3. Scalability

The system should be able to scale and expand easily in future, so that new features can be added to it later on.

Chapter 2

Literature Review

2.1 Related Research

2.1.1 Video Analysis and Data-driven Tactical Optimization of Sports Football Matches: Visual Recognition and Strategy Analysis Algorithm

2.1.1.1 Summary

Football paper [1], which describes an algorithmic method to evaluate football tactics by using realtime video analysis of the played games and an advanced mathematical model. For Object Detection **YOLOv5** is used which able to provide a greater than 92% level of accuracy by using sophisticated motion-based interpolation models and integrated with **Strategic Analysis (SA)** with **Markov Chain Models (MCM)**. The system monitors players and the ball, evaluates its trajectory and eventually determines tactical conclusions via a **Steady-State Analysis (SSA)** that forecasts long-term strategies. The dataset used is a football dataset from 6 top European leagues between 2014 and 2017 from **SoccerNet** — this provides a high quality streaming dynamic-view. This model scored 98.6% detection accuracy for player and ball tracking enabling use real time during match or post-match decision strategically.

2.1.1.2 Critical Analysis

While the paper introduces a sophisticated methodology for real-time tactical analysis, it presents several limitations. The approach depends heavily on the accuracy of YOLOv5, which, though advanced, may still struggle with occlusions leading to occasional inaccuracies in object detection. The reliance on predefined zones and a simplified Markov

model might oversimplify the complexity of dynamic football strategies. Furthermore, the evaluation is limited to certain European leagues, raising concerns about the generalizability of the findings across different playing styles or leagues. Future improvements could focus on incorporating more diverse datasets and refining the models to account for more detailed tactical behaviors.

2.1.1.3 Relationship to Soccerix

This research aligns with Soccerix’s goal of utilizing AI for precise ball and player detection and generating tactical insights. By applying similar advanced algorithms like YOLO and using SoccerNet dataset, our system may enhance its system’s ability to deliver actionable, data driven insights that improve team tactics and performance.

2.1.2 Tracking of Multi-Athlete and Action Recognition in Soccer Sports Video Using Deep Learning Techniques

2.1.2.1 Summary

The paper [5] focuses on automating the detection and recognition of key events in soccer games using deep learning models. The methodology includes pre-processing soccer video frames, extracting features using deep learning models like **ResNet152**, **Naive Bayes**, and **SVM**, and classifying events such as goals, tackles, and fouls. The **SoccerNet-v2** dataset, consisting of 550 soccer videos from top European leagues, was used to train and test the models. The proposed system is structured into data collection, pre-processing, feature extraction, and event prediction, achieving an accuracy of 88% with the ResNet152 model. The paper aims to reduce the time-consuming manual annotation process by automating event detection.

2.1.2.2 Critical Analysis

The paper’s strengths include the successful integration of multiple deep learning models, which improved action recognition accuracy, especially using concatenated pre-computed features. This approach enhanced performance, as seen in the ResNet152 model’s 88% accuracy. However, the reliance on precomputed features rather than raw video data limits flexibility and scalability. Additionally, the system’s dependence on a single dataset (SoccerNet-v2) could lead to overfitting, and the lack of real-time implementation testing raises concerns about its practical applicability.

2.1.2.3 Relationship to Soccerix

The research aligns with Soccerix as it uses deep learning models like ResNet152 to automate event detection, such as goals and fouls. This approach can be helpful for Soccerix's goal of using AI to track and analyze game events, providing accurate insights and enhancing tactical analysis.

2.1.3 Soccer Video Event Detection Using Metric Learning

2.1.3.1 Summary

The paper [3] proposes a solution for detecting events in soccer videos using **convolutional neural networks (CNNs)** and **gated recurrent units (GRUs)**. This makes use of **Siamese networks** that uses **contrastive loss** in detecting different soccer events such as goals, fouls, penalty and corner kicks. In this work we take as reference the **SoccerEvent** and **SoccerNet** datasets. The approach attains 64.9% mAP on SoccerNet and 93.76% mAP in the SoccerEvent data set. The best-fit architecture in our case was EfficientNetB0 and GRU. This paper demonstrates the effectiveness of metric learning for event detection, showing that model trained by pair-wise ranking can learn a more general feature and perform cross-match generalization over diverse soccer game data.

2.1.3.2 Critical Analysis

This paper is to fully utilize the spatial and temporal information from soccer videos by using Siamese networks within a deep learning architecture for CNN and GRU. Moreover, by applying contrastive loss, the event detection accuracy can be further boosted to 93.76% mAP on SoccerEvent dataset. In addition, collecting a diverse dataset also helps the model to generalize better across different conditions such as stadiums, teams etc. Despite this, the model records a relatively lower performance (64.9%) on SoccerNet dataset that could be an indication of shortfalls in dealing with certain events or datasets with less diversity. Furthermore, the system's reliance on computationally intensive architectures like EfficientNetB0 may limit its scalability and real-time application. Expanding the evaluation across more diverse soccer datasets or exploring lightweight architectures for real-time analysis could address these gaps.

2.1.3.3 Relationship to Soccerix

The paper is relevant to Soccerix as it combines CNNs and GRUs to accurately detect events like goals and corner kicks. Use of these advanced techniques might enhance Soc-

cerix’s system by improving event detection and providing detailed, data-driven insights into game dynamics.

2.1.4 RMS-Net: Regression and Masking for Soccer Event Spotting

2.1.4.1 Summary

The paper [8] introduces a lightweight network for action spotting in soccer videos, focusing on detecting key events such as goals, substitutions, and cards. The methodology involves predicting both the event class and its temporal offset using **2D convolutional features with a masking strategy** to handle ambiguous frames. The system is tested on the **SoccerNet dataset**, containing 500 full soccer matches, and achieves a 65.5% Average-mAP on the test set, surpassing prior methods. The model integrates **temporal convolutions** and **regression** strategies to improve temporal precision. Two techniques, **data balancing** and **masking** of irrelevant frames, are incorporated to mitigate class imbalance and enhance feature extraction.

2.1.4.2 Critical Analysis

RMS-Net excels in its lightweight design and modularity, allowing for easy integration with existing models while addressing event sparsity in soccer videos. Its innovative use of temporal regression improves the precision of event spotting, outperforming prior baselines. However, the model struggles with detecting subtle events like cards due to limited visual cues, and the masking strategy, while effective, may reduce context, leading to misclassifications in complex scenarios. Moreover, the system heavily relies on pre-extracted features, limiting its adaptability to more dynamic contexts like player interactions. Further improvements could focus on enhancing recognition of minor events and reducing reliance on specific pre-trained models.

2.1.4.3 Relationship to Soccerix

The paper’s approach to detecting key soccer events using temporal features and masking aligns with Soccerix’s aim to capture and analyze events like goals and free kicks. The method’s emphasis on accuracy and event detection efficiency directly supports our system’s goal of providing data-driven insights to improve team tactics and gameplay decisions.

2.1.5 Extraction of Positional Player Data from Broadcast Soccer Videos

2.1.5.1 Summary

The paper [7] presents a modular pipeline for fully-automated extraction of positional player data from broadcast soccer videos. The system includes several key components: sports field registration, player detection, team assignment, and position estimation. The approach leverages **homography matrices** to transform player positions from image to field coordinates. The system was evaluated on multiple datasets, including broadcast videos from the **German Bundesliga**, and achieved promising results, with the mean position error being less than 2 meters for most frames. Models like **Faster R-CNN** and **CenterTrack** are employed for player detection. The pipeline's modularity enables individual evaluation of each sub-task, with the ultimate goal of assisting in tactical analysis and player performance evaluation. Player detection achieved average precision of 92.8% on the Soccer Player dataset and 90.1% on the ISSIA-CNR dataset.

2.1.5.2 Critical Analysis

The pipeline's strengths include its generalizability to new datasets, modular design, and robust evaluation across different camera settings. The use of homography matrices allows accurate mapping of player positions from video frames to field coordinates. However, the system's modular approach, while beneficial for flexibility, may compound errors between sub-tasks, reducing overall accuracy. Additionally, the system's reliance on visual input makes it vulnerable to inaccuracies caused by poor video resolution or lighting. Future work could focus on improving player tracking and addressing occlusion issues for more consistent results.

2.1.5.3 Relationship to Soccerix

The paper's pipeline for extracting player positions from broadcast videos, using models like Faster R-CNN and homography matrices, aligns with Soccerix's AI-based player and ball tracking. It also supports key Soccerix goals like event detection and tactical analysis, offering a solid foundation for data-driven insights in soccer.

2.2 Analysis Summary of research items

ID	Title	Methodology	Dataset	Evaluation	Limitation
1	Video analysis and data-driven tactical optimization of sports football matches: Visual recognition and strategy analysis algorithm (2024) [1]	YOLOv5, Markov Chain Models (MCM)	SoccerNet	Detection accuracy of 98.6%	Reliance on YOLOv5, and predefined zone and simplified Markov Model
2	Tracking of Multi-Athlete and Action Recognition in Soccer Sports Video Using Deep Learning Techniques (2024) [5]	ResNet152, Naive Bayes, and SVM	SoccerNet-v2	Event recognition accuracy of 88%	Reliance on pre-computed features rather than raw data
3	Soccer Video Event Detection Using Metric Learning (2022) [3]	convolutional neural networks (CNNs), gated recurrent units (GRUs), Siamese networks and contrastive loss	SoccerEvent and SoccerNet	64.9% mAP on SoccerNet and 93.76% mAP in the SoccerEvent dataset	Computationally intensive architectures (EfficientNetB0) limit its scalability and lower performance (64.9%) on SoccerNet dataset
4	RMS-Net: Regression and Masking for Soccer Event Spotting (2021) [8]	2D convolutional features with a masking strategy, temporal convolutions and regression, data balancing and masking of irrelevant frames	SoccerNet	Average mAP of 65.5%	Detecting subtle events like cards, heavily relies on pre-extracted features, limiting its adaptability to more dynamic contexts

ID	Title	Methodology	Dataset	Evaluation	Limitation
5	Extraction of Positional Player Data from Broadcast Soccer Videos (2022) [7]	Homography matrices, Faster R-CNN, Center-Track	Multiple datasets, including broadcast videos from the German Bundesliga and ISSIA-CNR	Average precision of 92.8% for player detection on the Soccer Player dataset and 90.1% on the ISSIA-CNR dataset	Modular approach may compound errors between sub-tasks reducing overall accuracy

Table 2.1: Literature Review Table

Table 2.1 presents the comprehensive literature review.

Chapter 3

Proposed Approach

The dataset used for **Soccerix** is sourced from **SoccerNet** [6]. For **event detection**, the SoccerNet Action Spotting dataset includes 500 soccer broadcast videos and covers 17 event categories: Penalty, Kick-off, Goal, Substitution, Offside, Shots on target, Shots off target, Clearance, Ball out of play, Throw-in, Foul, Indirect free-kick, Direct free-kick, Corner, Yellow card, Red card, and Yellow->red card. For **object detection and tracking**, the SoccerNet Tracking dataset includes classes such as player team left, player team right, goalkeeper team left, goalkeeper team right, main referee, side referee, staff, and ball, based on 100 video clips of 30 seconds each at 1080p resolution.

The **Soccerix** system begins by analyzing video footage, detecting key events like goals, free kicks, and corner kicks while using object detection to identify players, ball, goalkeepers and referees on the field. It tracks object movements, assigns players to teams, and analyzes team formations. Soccer field key points are detected for creating a bird's-eye view of the field using perspective transformation, highlighting player positions. The detected events are linked to specific teams, providing contextual insights. Finally, this data is transformed into detailed analysis, automated reports, and visualizations, offering coaches and analysts a comprehensive understanding of the match through dashboards that present actionable metrics and performance evaluations.

Below is a detailed outline of the tentative proposed approach for each feature in Soccerix. This includes various potential models and techniques that may be employed throughout the project. We will explore these options while also considering additional alternatives as needed to adapt to any challenges that may arise.

3.0.1 Video Input Processing

The system breaks down the raw input video into individual frames to analyze specific moments in the game. Each subsequent module applies preprocessing for its task, such as

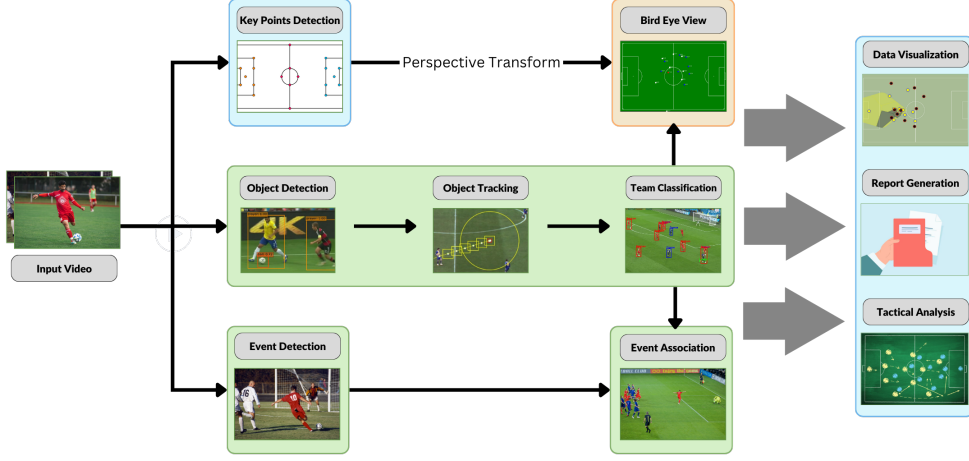


Figure 3.1: Our proposed architecture for soccer tactical analysis. We process video data to detect events, track players, and generate team insights, leading to automated report generation and visualizations for post-match review.

resizing images to enhance the visibility of the ball and players for object detection and tracking.

3.0.2 Object Detection

In the Soccerix system, object detection models like YOLO or Fast R-CNN may be used to identify players, goalkeepers, referees and the ball on the field, recognizing and localizing them within a frame. YOLO’s loss function, for example, includes several components, such as the localization error, confidence score error, and classification error, and is represented as:

3.0.2.1 Yolo Loss Function:

The YOLO loss function consists of three main components:

1. Localization loss
2. Confidence loss
3. Classification loss

$$L = \lambda_{\text{coord}} \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{I}_{ij}^{\text{obj}} \left[(x_i - \hat{x}_i)^2 + (y_i - \hat{y}_i)^2 + (\sqrt{w_i} - \sqrt{\hat{w}_i})^2 + (\sqrt{h_i} - \sqrt{\hat{h}_i})^2 \right] \quad (3.1)$$

$$+ \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{K}_{ij}^{\text{obj}} (C_i - \hat{C}_i)^2 + \lambda_{\text{noobj}} \sum_{i=0}^{S^2} \sum_{j=0}^B \mathbb{K}_{ij}^{\text{noobj}} (C_i - \hat{C}_i)^2 \quad (3.2)$$

$$+ \sum_{i=0}^{S^2} \mathbb{K}_i^{\text{obj}} \sum_{c \in \text{classes}} (p_i(c) - \hat{p}_i(c))^2 \quad (3.3)$$

Explanation:

- S^2 : Number of grid cells.
- B : Number of bounding boxes per grid cell.
- λ_{coord} : Weight for the localization (coordination) loss.
- λ_{noobj} : Weight for the no-object confidence loss.
- x_i, y_i, w_i, h_i : The predicted bounding box parameters (center coordinates, width, height).
- $\hat{x}_i, \hat{y}_i, \hat{w}_i, \hat{h}_i$: The ground truth bounding box parameters.
- C_i : Confidence score for the predicted bounding box.
- $\hat{C}_i$: Ground truth confidence score.
- $p_i(c)$: Predicted probability of class c .
- $\hat{p}_i(c)$: Ground truth class probability.

3.0.3 Object Tracking

To track the object movements throughout the match, algorithms such as ByteTrack, FairMOT, or DeepSORT might be used to associate detected objects across frames. The combined process of object detection and tracking enables the system to gather positional and movement data, which is essential for the analysis of gameplay dynamics and team performance. High level architecture of ByteTrack is shown in figure 3.2.

3.0.4 Event Detection

Event detection in Soccerix is inspired by this paper [2], which focuses on recognizing key events like goals, free kicks, and corner kicks. The process begins by taking video input and converting it into individual frames, which are then passed through a Variational

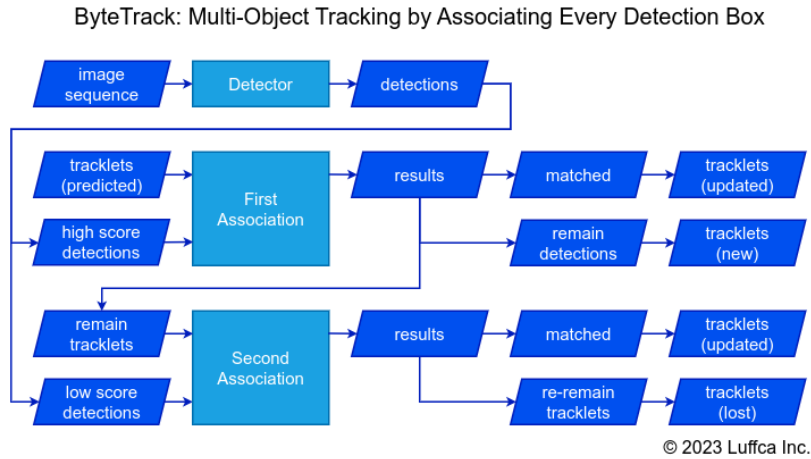


Figure 3.2: ByteTrack: A multi-object tracking framework [4] utilizing a two-stage association to handle all detection boxes, including low-confidence ones.

Autoencoder (VAE) for initial filtering to identify potential event frames and remove non-highlight frames. The filtered event frames are subsequently fed into a classifier network comprising architectures like ResNet, VGG16, InceptionResNetV2, and EfficientNetB0, which are responsible for accurately categorizing the events. This pipeline ensures that only relevant event frames are processed further, significantly enhancing the precision of event detection while reducing computational overhead by discarding irrelevant frames early in the pipeline.

Other models for temporal action detection can also be employed. Approaches like Temporal Convolutional Networks (TCN) or Recurrent Neural Networks (RNN), particularly Long Short-Term Memory (LSTM), are commonly used for this task. These models analyze sequences of frames over time to detect specific events such as goals, free kicks, or free-kicks.

3.0.5 Bird's Eye View

This module or feature can be divided into the following four sub-tasks:

1. **Soccer pitch Keypoints and lines detection:**

Detect key pitch elements such as goalposts, center circle, penalty spots, sidelines, and pitch lines using techniques like Hough Transform and semantic segmentation, possibly employing models like HRNet and YOLO for key point detection.

2. **Homography & perspective transformation:**

Apply homography transformation to map the detected key points from the 2D camera view to a top-down (bird's-eye) view, correcting for camera perspective.

3. Bird's eye view:

Create a top-down view of the soccer field using the homography transformation, enabling precise tracking of player positions and movements for tactical analysis.

3.0.6 Team Classification

Team classification refers to the process of grouping individuals into distinct teams based on visual data. Various models can be used to generate embedding vectors for players using contrastive learning, allowing the model to create meaningful representations of players based on their visual features. If the learned representations are of significantly higher dimensions, dimensionality reduction techniques can be employed. Models like **autoencoders** combine these two steps into one. Afterwards, clustering can be performed to classify the players into their respective teams.

3.0.6.1 Models for Representation Learning:

- **Sigmoid Loss for Image and Language Pre-training (SigLIP):** This model generates image embeddings using a contrastive learning framework. The vision model from SigLIP, without any head or projection on top, generates image embeddings. It focuses on learning robust representations from visual data without any text input.
- **Siamese Networks:** Used to learn embeddings for images by comparing pairs of images and training the network to minimize the distance between similar images and maximize the distance between dissimilar images.

The contrastive loss function used in Siamese Networks can be represented as:

$$L(y, \mathbf{f}(x_1), \mathbf{f}(x_2)) = (1 - y) \cdot \frac{1}{2} \|\mathbf{f}(x_1) - \mathbf{f}(x_2)\|^2 + y \cdot \frac{1}{2} \max(0, m - \|\mathbf{f}(x_1) - \mathbf{f}(x_2)\|)^2 \quad (3.4)$$

Where:

- y is a binary label indicating whether the pair (x_1, x_2) are similar (1) or dissimilar (0).
- $\mathbf{f}(x)$ is the embedding function output by the Siamese network.
- $\|\cdot\|$ denotes the Euclidean distance between the embeddings.
- m is the margin that defines the minimum distance between dissimilar pairs.

3.0.6.2 Techniques for Dimensionality Reduction:

- **Uniform Manifold Approximation and Projection (UMAP):** A popular technique for dimensionality reduction that preserves the local structure of the data.
- **Principal Component Analysis (PCA):** A statistical technique that transforms high-dimensional data into a lower-dimensional form while preserving as much variance as possible.

3.0.6.3 Clustering Algorithms For Team Classification:

- **K-Means Clustering:** A technique that is used to reduce the dimensionality of data while preserving the variance of the data.

The K-Means algorithm minimizes the following objective function:

$$J = \sum_{k=1}^K \sum_{x_i \in C_k} ||x_i - \mu_k||^2 \quad (3.5)$$

Where:

- K is the number of clusters.
 - C_k is the set of points in cluster k .
 - μ_k is the centroid of cluster k .
- **Density-Based Spatial Clustering of Applications with Noise (DBSCAN):** A clustering algorithm that groups data points into clusters based on density. It identifies dense regions of data points and assigns points within these regions to the same cluster.

3.0.7 Event Association

Event association links detected events to specific teams. This process typically relies on custom algorithms that analyze player positions and ball trajectories. Rule-based systems, using domain-specific rules, are combined with machine learning techniques for events like goals, free-kicks and corner kicks.

3.0.8 Team Tactical Analysis

This module may be done by writing custom algorithms to analyze video data. This allows us to analyze things like how the teams were positioned at certain event like goals and how many goals happened in the game, indicating how well each team did overall.

3.0.9 Report Generation

The system creates clear and concise reports that highlight important information like team composition, event frequency, and how well each team performed. These reports are generated automatically using advanced technology that can transform data into easy-to-understand language. Natural Language Processing (NLP) models like T5, GPT-2, or BERT are used to convert the data into human-readable text.

3.0.10 Visualizations

Once the data has been processed, visualizations like team pitch control diagrams and ball trajectory maps will be created to bring the game to life. Interactive dashboards will present this tactical analysis in an easy-to-understand format, giving coaches and analysts a clear picture of team performance and match dynamics.

Chapter 4

Conclusions and Future Work

4.1 Conclusion

Our AI-driven football analysis tool, **Soccerix**, has the potential to transform tactical analysis after a match. Utilising modern data-driven methods, we're turning unprocessed video into insightful knowledge. Soccerix keeps tabs on player movements, identifies important events, and produces comprehensive tactical reports. To effectively analyse gameplay dynamics, generate a birds-eye view of player positions, and accurately classify team performance, our team has experimented with a variety of models, methodologies, and strategies. The interactive dashboards of the application provide coaches and analysts with significant performance indicators and insights, enabling them to make informed decisions based on data and improve their team's tactics.

4.2 Future Work

Future developments for **Soccerix** will focus on enhancing event detection and analysis through advanced algorithms and incorporating 3D reconstruction of games for more detailed spatial insights. We aim to integrate predictive analytics to forecast optimal performance strategies and implement real-time inferencing to provide immediate feedback during matches. These advancements will further refine tactical analysis, offering teams a cutting-edge tool for continuous performance optimization.

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